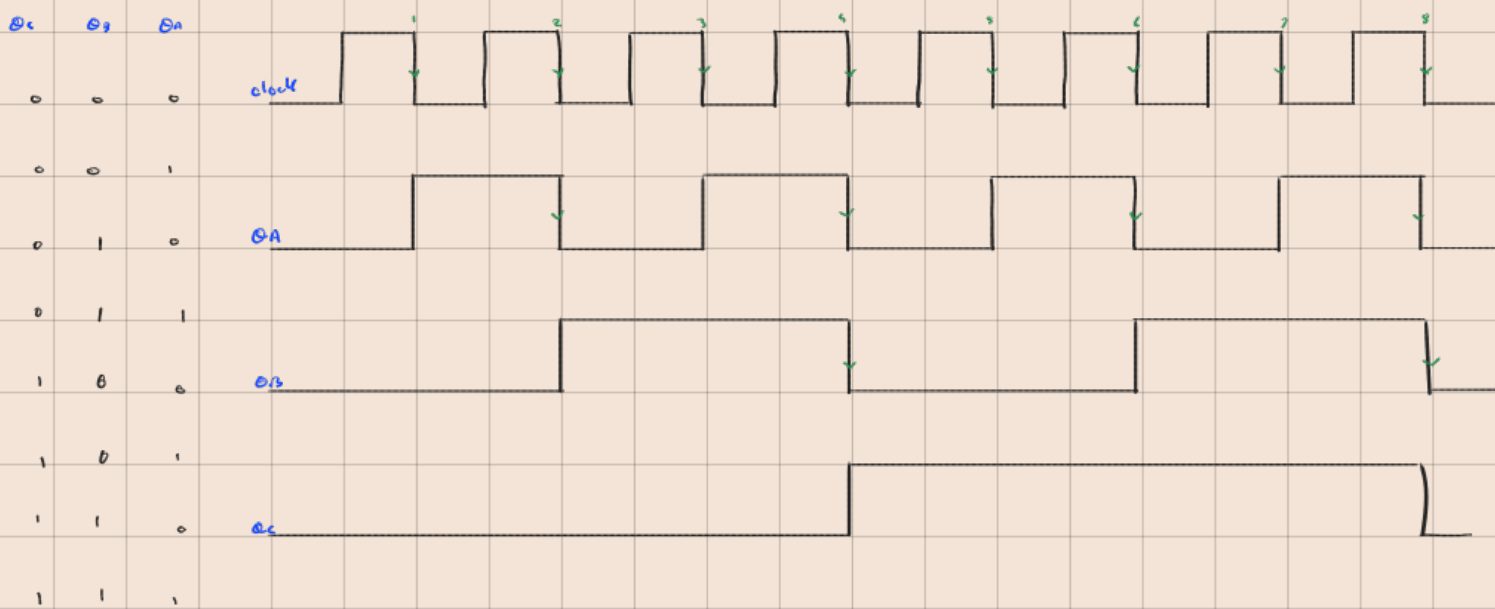
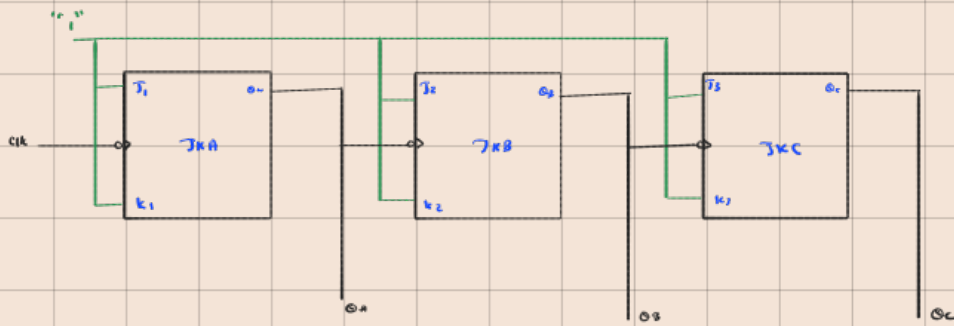
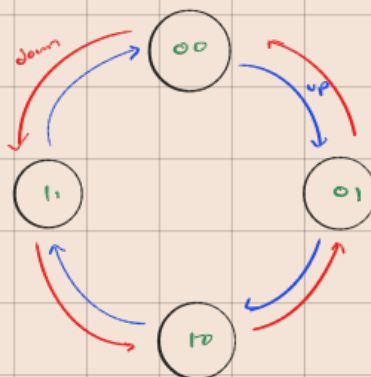


3 Bit (Using JK Fliplop)



State diagram for a 2 bit for simplicity



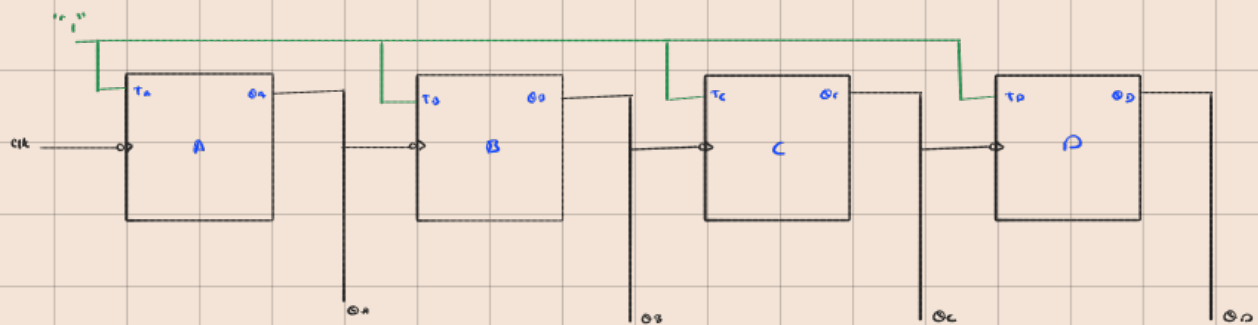
4 Bit (Using T Flip Flops)

∴ In order to make an asynchronous counter we just need the toggling functionality

- for Jk condition is to set both pins to "1"
- for T Flip Flops set input to "1"
- for D Flip Flops connect the inverted output back to input.

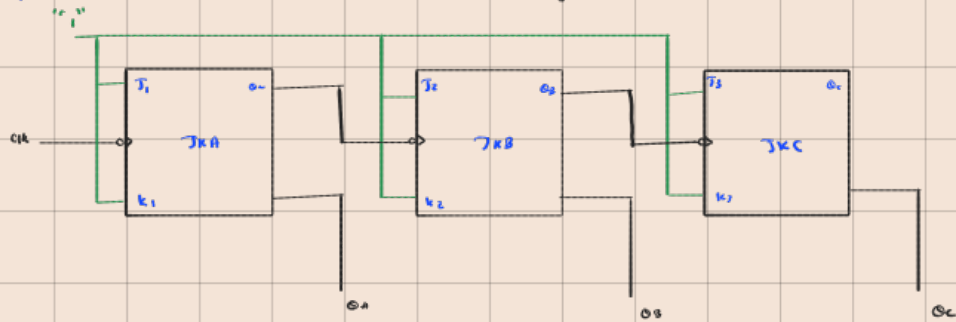
- what if it's a mod counter

- Connect And or NAND gate to the outputs of Flip Flops for a certain output.
- when that output is true the gate outputs a signal to clear / Reset pins of all the Flip Flops.

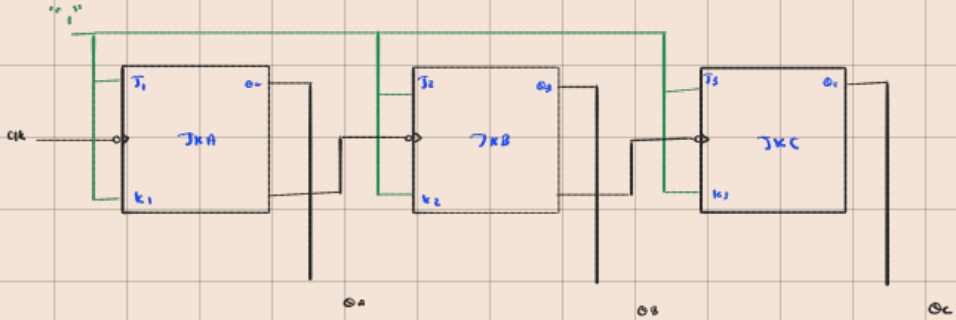


Down counter logic

Logic 1: (use the inverted outputs, clock come from normal)

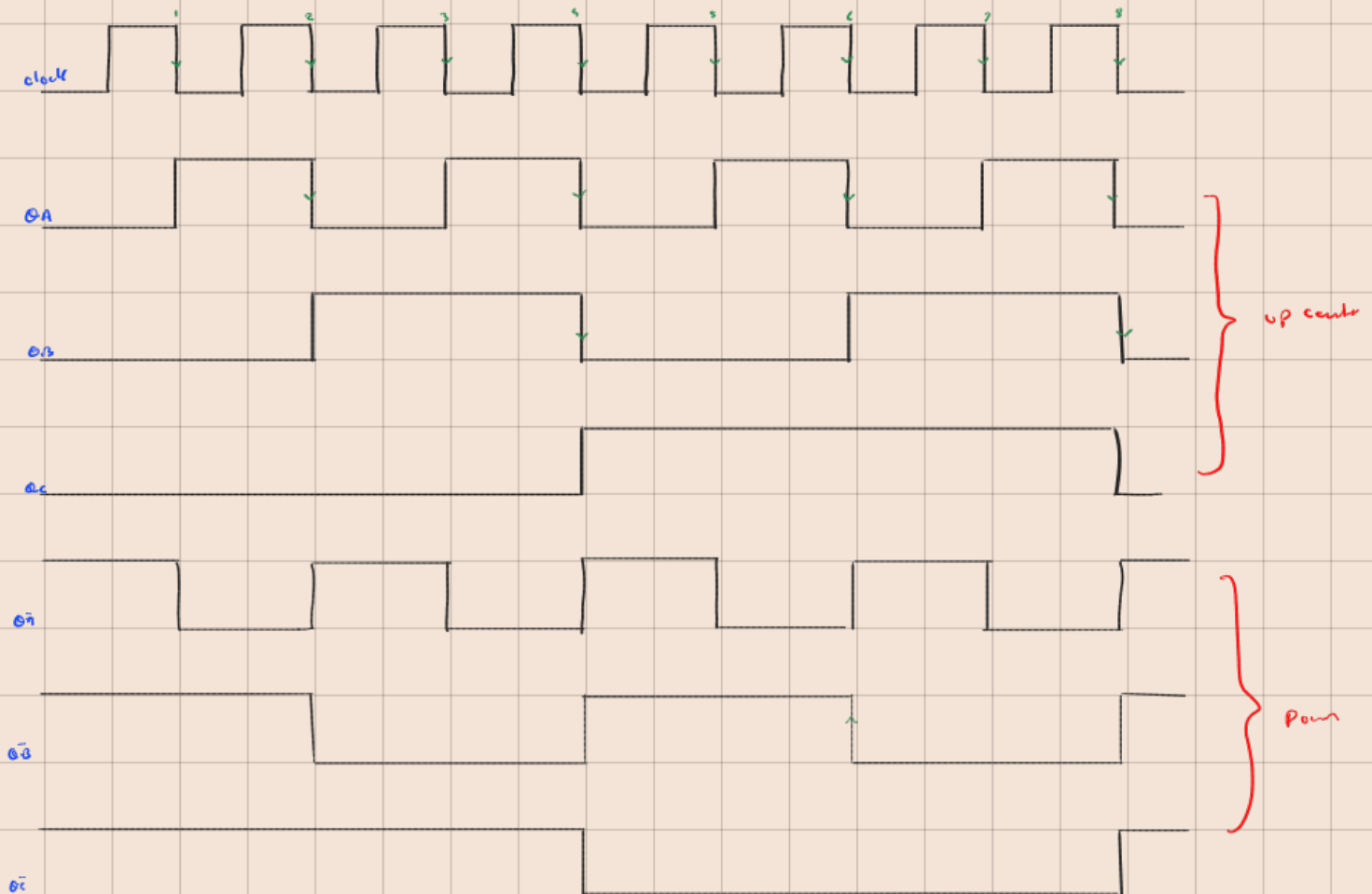


Logic 1: (use the inverted outputs, clock come from normal)



Down Counter

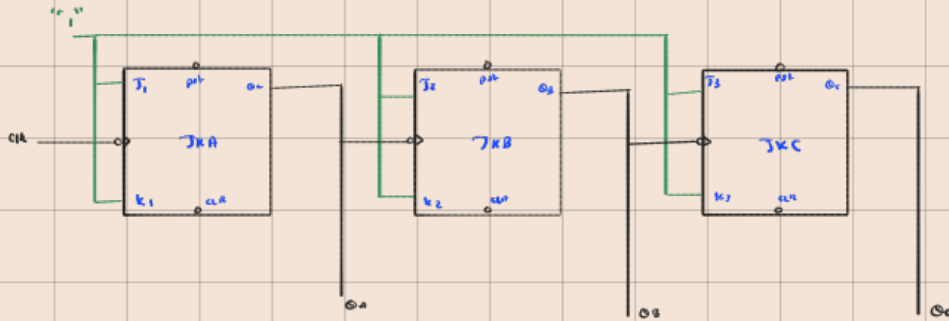
QC	QB	QA
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0
1	1	1



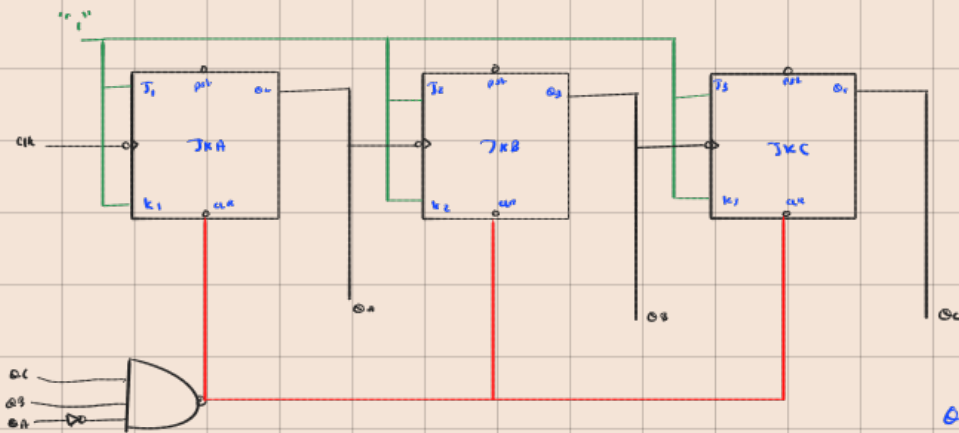
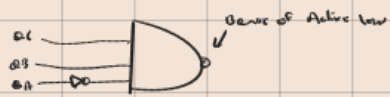
up/ Down Counter.

Mod Counter

Reset/Clr Pins : Active low



Mod 6: (0 1 0)



	Q1	Q2	Q3
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	0	0	0

Reset

Decade Ripple Counter (BCD Counter)

$\begin{matrix} Q_D & Q_C & Q_B & Q_A \\ (1 & 0 & 1 & 0) \end{matrix} \rightarrow \text{Resets.}$

Negative edge $\rightarrow \bar{Q}$ is clock $\rightarrow$ Up

Positive edge $\rightarrow \bar{Q}$ is clock $\rightarrow$ Up

Negative edge $\rightarrow \bar{Q}$ is clock $\rightarrow$ Down

Positive edge $\rightarrow Q$ is clock $\rightarrow$ Up

